Lathi 20题练习册 第9批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

脉冲信号的能量及幅度、时移影响

Lathi 1.1-1 | 原书第 136 页

Boulet 第1章：Finite-Energy and Finite-Power Signals， p.21；Transformations of the Time Variable， p.4

原图P1.1-1(a)-(d)四个信号均已完整附上。

1.1-1 Find the energies of the signals illustrated in Fig. P1.1-1. Comment on the effect on energy of sign change, time shifting, or doubling of the signal. What is the effect on the energy if the signal is multiplied by k?

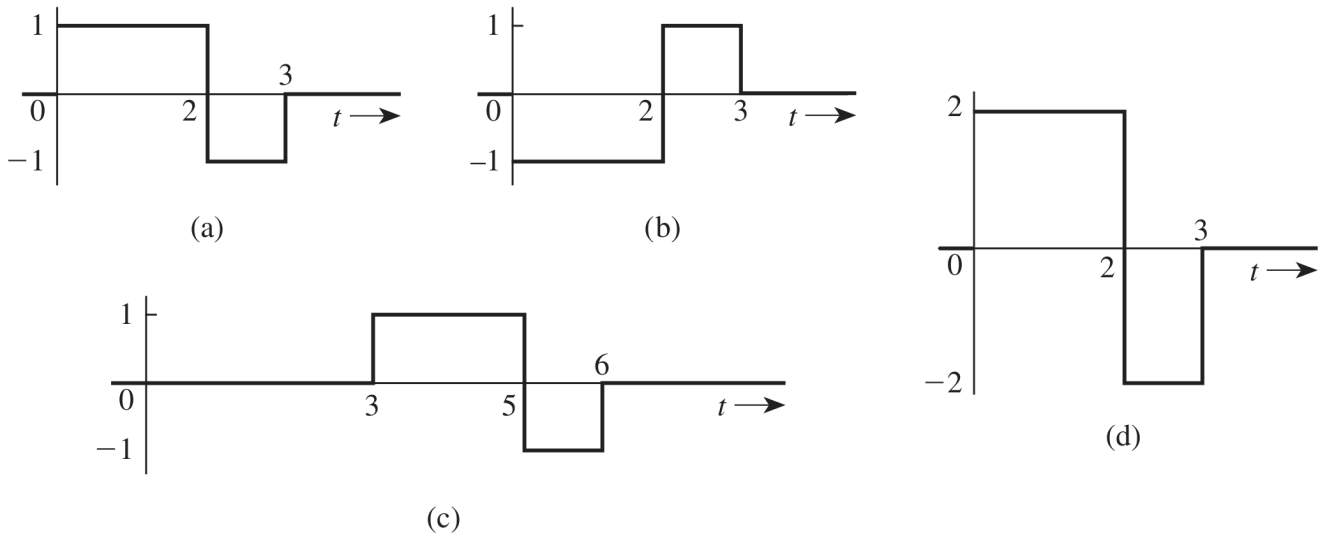


Figure P1.1-1

演算区

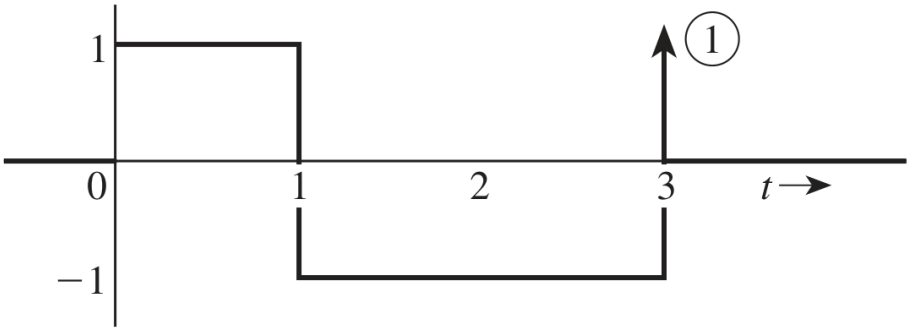
含冲激的分段信号积分与作图

Lathi 1.4-9 | 原书第 142 页

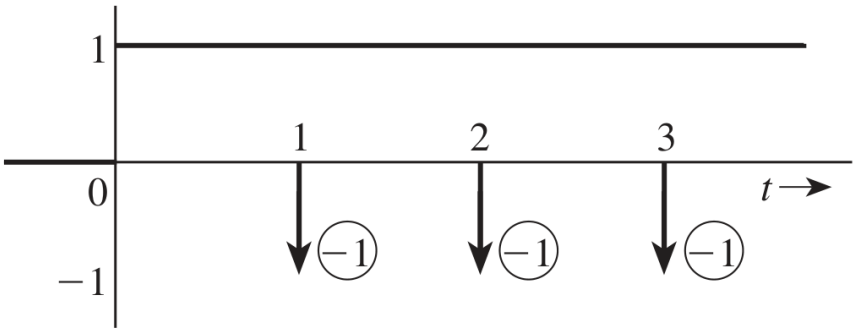
Boulet 第1章：Generalized Functions，p.26

原图P1.4-9(a)、(b)均已附，圆圈数字为冲激权重。

1.4-9 Find and sketch $\int_{-\infty}^{t} x(t)\,dt$ for the signals $x(t)$ illustrated in Fig. P1.4-9.



(a)



(b)

Figure P1.4-9

演算区

非周期信号奇分量的周期性

Lathi 1.5-5 | 原书第 143 页

Boulet 第1章：Even and Odd Signals, p.23；Periodic Signals, p.8

1.5-5 An aperiodic signal is defined as $x(t) = \sin(\pi t)u(t)$, where $u(t)$ is the continuous-time step function. Is the odd portion of this signal, $x_o(t)$, periodic? Justify your answer.

演算区

由输入导数选择支路的系统性质

Lathi 1.7-6 | 原书第 145 页

Boulet 第1章：System Models and Basic Properties，p.34

展开所引用1.7-4的要求：分别判断系统是否(a)BIBO稳定、(b)因果、(c)可逆、(d)线性、(e)无记忆、(f)时不变；每项回答是或否，并给出数学论证。

1.7-6 Repeat Prob. 1.7-4 for a system with input $x(t)$ that produces output $y(t)$ according to

$$y(t-1) = \begin{cases} x(t-1) & \text{when } \frac{d}{dt}x(t) \geq 0 \\ x(t-2) & \text{otherwise} \end{cases}$$

演算区

离散序列的反转、压缩、扩展与原点定位

Lathi 3.2-3 | 原书第 316 页

Boulet 第1章：Transformations of the Time Variable, p.4

保留原书换行处的n=0箭头；按原书离散时间扩展约定，下标非整数的位置补零。

3.2-3 Letting $\downarrow$ identify $n = 0$, define the nonzero values of signal $x[n]$ as $[-1, 2, -3, 4, -5, 4, -3, \underset{\downarrow}{2}, -1]$.

(a) Using vector form, represent signal $y[n] = x[-3n + 2]$. Be sure to identify the $n = 0$ element.

(b) Using vector form, represent signal $z[n] = x[n/2 - 3]$. Be sure to identify the $n = 0$ element.

演算区

斜坡脉冲与其压缩版本的卷积

Lathi 2.4-35 | 原书第 231 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-35 An analog signal is given by $x(t) = t[u(t) - u(t-1)]$, as shown in Fig. P2.4-35. Determine and plot $y(t) = x(t) * x(2t)$.

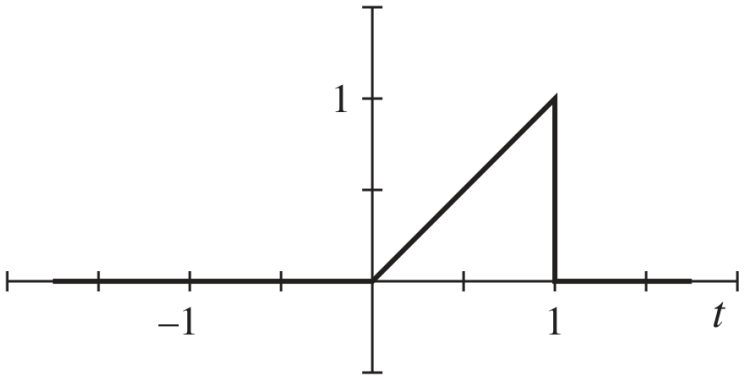


Figure P2.4-35

演算区

双边指数冲激响应的因果性与输出

Lathi 2.4-40 | 原书第 231 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

2.4-40 An LTIC system has impulse response $h(t) = 3e^{-|t|}$.

(a) Is the system causal? Mathematically justify your answer.

(b) Determine the zero-state response of this system if the input is $x(t) = u(2 - t)$.

演算区

积分器及其并联、串联冲激响应

Lathi 2.4-42 | 原书第 232 页

Boulet 第2章：Properties of Linear Time-Invariant Systems, p.74

2.4-42 Consider an integrator system, $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$.

(a) What is the unit impulse response $h_{\mathrm{i}}(t)$ of this system?

(b) If two such integrators are put in parallel, what is the resulting impulse response $h_{\mathrm{p}}(t)$?

(c) If two such integrators are put in series, what is the resulting impulse response $h_{\mathrm{s}}(t)$?

演算区

复值LTI系统的卷积与叠加响应

Lathi 2.4-48 | 原书第 233 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

原书(c)写引用part(a)，但(a)仅问因果性；结合题意，该处疑应指(b)所得y1，再利用线性与时不变性质。截图保留原文。

2.4-48 A system is called complex if a real-valued input can produce a complex-valued output. Suppose a linear time-invariant complex system has impulse response $h(t) = j[u(-t+2) - u(-t)]$.

(a) Is this system causal? Explain.

(b) Use convolution to determine the zero-state response $y_1(t)$ of this system in response to the unit-duration pulse $x_1(t) = u(t) - u(t-1)$.

(c) Using the result from part (a), determine the zero-state response $y_2(t)$ in response to $x_2(t) = 2u(t-1) - u(t-2) - u(t-3)$.

演算区

指数正弦离散冲激响应的查表卷积

Lathi 3.8-9 | 原书第 322、285 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54

附原题要求使用的表3.1表头及第10行参考公式，未代入本题参数求解；表中相角phi应注意对应复数的象限。

3.8-9 Find the (zero-state) response $y[n]$ of an LTID system if its input $x[n] = (2)^n u[n-1]$, and

$$h[n] = (3)^n \cos\left(\frac{\pi}{3}n - 0.5\right) u[n]$$

Find your answer using only Table 3.1.

TABLE 3.1 Select Convolution Sums

No.	$x_1[n]$	$x_2[n]$	$x_1[n] * x_2[n] = x_2[n] * x_1[n]$
10	$\lvert\gamma_1\rvert^n \cos(\beta n + \theta) u[n]$	$\lvert\gamma_2\rvert^n u[n]$	$\frac{1}{R}[\lvert\gamma_1\rvert^{n+1} \cos[\beta(n+1)+\theta-\phi] - \lvert\gamma_2\rvert^{n+1} \cos(\theta-\phi)]u[n]$ $R = \left[\lvert\gamma_1\rvert^2 + \lvert\gamma_2\rvert^2 - 2\lvert\gamma_1\rvert\lvert\gamma_2\rvert \cos\beta\right]^{1/2}$ $\phi = \tan^{-1}\left[\frac{(\lvert\gamma_1\rvert \sin\beta)}{(\lvert\gamma_1\rvert \cos\beta - \lvert\gamma_2\rvert)}\right]$

演算区

三阶连续系统的特征模态与初值响应

Lathi 2.2-10 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

展开所引用2.2-4的要求：(a)求特征多项式、特征方程、特征根和特征模态；(b)按给定三个初值求 t>=0 的零输入响应。

2.2-10 Repeat Prob. 2.2-4 for

$$(D+1)(D^2+5D+6)y(t) = Dx(t)$$

$$\text{with } y_0(0^-) = 2,\ \dot{y}_0(0^-) = -1, \text{ and } \ddot{y}_0(0^-) = 5.$$

演算区

LC电路的建模、自然模态与完整响应

Lathi 2.4-32 | 原书第 228、229、230 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

原题跨页，(a)-(e)及图P2.4-32均已附；(c)的一般参数与(e)给定L=C=1的条件分别按原题使用。

2.4-32 Consider the electric circuit shown in Fig. P2.4-32.

(a) Determine the differential equation that relates the input $x(t)$ to output $y(t)$. Recall that $i_C(t) = C\frac{dv_C(t)}{dt}$ and $v_L(t) = L\frac{di_L(t)}{dt}$.

(b) Find the characteristic equation for this circuit, and express the root(s) of the characteristic equation in terms of L and C.

(c) Determine the zero-input response given an initial capacitor voltage of one volt and an initial inductor current of zero amps. That is, find $y_0(t)$ given $v_C(0) = 1$ V and $i_L(0) = 0$ A. [*Hint:* The coefficient(s) in $y_0(t)$ are independent of L and C.]

(d) Plot $y_0(t)$ for $t \geq 0$. Does the zero-input response, which is caused solely by initial conditions, ever "die out"?

(e) Determine the total response $y(t)$ to the input $x(t) = e^{-t}u(t)$. Assume an initial inductor current of $i_L(0^-) = 0$ A, an initial capacitor voltage of $v_C(0^-) = 1$ V, $L = 1$ H, and $C = 1$ F.

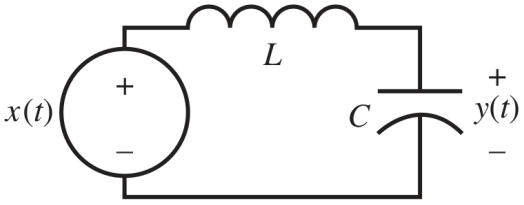


Figure P2.4-32

含出生、死亡与移民的人口差分模型

Lathi 3.4-2 | 原书第 317 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations, p.96

3.4-2 Let $p[n]$ be the population of a certain country at the beginning of the nth year. The birth and death rates of the population during any year are 3.3 and 1.3%, respectively. If $i[n]$ is the total number of immigrants entering the country during the nth year, write the difference equation relating $p[n+1]$, $p[n]$, and $i[n]$. Assume that the immigrants enter the country throughout the year at a uniform rate.

演算区

数字积分器的阶跃响应及连续极限

Lathi 3.4-4 | 原书第 317 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，p.96

所引用例3.9的系统方程已在原题中完整给出，无需另读例题演算。

3.4-4 The digital integrator in Ex. 3.9 is specified by

$$y[n] - y[n-1] = Tx[n]$$

If an input $u[n]$ is applied to such an integrator, show that the output is $(n+1)Tu[n]$, which approaches the desired ramp $nTu[n]$ as $T \to 0$.

演算区

离散系统的特征模态与实值共振输入

Lathi 3.10-2 | 原书第 327 页

Boulet 第3章：Eigenfunctions of LTI Difference and Differential Systems，p.117；Characteristic Polynomials and Stability，p.112

3.10-2 If one exists, determine a real input $x[n]$ that will cause resonance in the causal LTID system described by $(E^2+1)\{y[n]\}=(E+0.5)\{x[n]\}$. If no such input exists, explain why not.

演算区

（复习）三角与复指数傅里叶级数及幅相线谱转换

Lathi 6.3-3 | 原书第 672 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal，pp.134-136；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137

复习题：此前见第2批。复习第2批：重练同频三角项的合并、幅相线谱以及三角级数与复指数级数的互相转换。已重新核对Lathi原书，完整保留给定信号及(a)至(d)；原式三项均含相同的角频率，不能误读成不同谐波。无外部图或公式依赖。

6.3-3 A periodic signal $x(t)$ is expressed by the following Fourier series:

$$x(t) = 3\cos t + \sin\left(t - \frac{\pi}{6}\right) - 2\cos\left(t - \frac{\pi}{3}\right)$$

(a) Sketch the amplitude and phase spectra for the trigonometric series.
(b) By inspection of spectra in part (a), sketch the exponential Fourier series spectra.
(c) By inspection of spectra in part (b), write the exponential Fourier series for $x(t)$.
(d) Show that the series found in part (c) is equivalent to the trigonometric series for $x(t)$.

演算区

（复习）由复指数傅里叶级数求三角线谱与带宽

Lathi 6.3-5 | 原书第 672、673 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal，pp.134-136；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137

复习题：此前见第2批。复习第2批：重练从复系数读出双边线谱，再转换为紧凑三角级数并判断带宽。已重新核对Lathi原书，保留p.672末尾的完整给定级数及p.673开头的(a)至(d)，避免跨页漏题；无需附加图。

6.3-5 The exponential Fourier series of a certain function is given as

$$x(t) = (2+j2)e^{-j3t} + j2e^{-jt} + 3 - j2e^{jt} + (2-j2)e^{j3t}$$

(a) Sketch the exponential Fourier spectra.
(b) By inspection of the spectra in part (a), sketch the trigonometric Fourier spectra for $x(t)$. Find the compact trigonometric Fourier series from these spectra.
(c) Show that the trigonometric series found in part (b) is equivalent to the exponential series for $x(t)$.
(d) Find the signal bandwidth.

演算区

（复习）周期信号时移与时间尺度变换的傅里叶级数性质

Lathi 6.3-10 | 原书第 674 页

Boulet 第4章：Properties of Continuous-Time Fourier Series，pp.139-140（Time Shifting、Time Scaling）

复习题：此前见第2批。复习第2批：重练时移与时间尺度变换对级数系数、幅相线谱及谐波频率位置的影响。已重新核对Lathi原书，完整保留左栏题干、(a)和右栏(b)及其直观解释要求；原题列出的关系式属于要求证明的结论，并非另附解答。图P6.3-9属于上一题，本题不依赖该图。

6.3-10 A periodic signal $x(t)$ is expressed as an exponential Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

(a) Show that the exponential Fourier series for $\hat{x}(t) = x(t - T)$ is given by

$$\hat{x}(t) = \sum_{n=-\infty}^{\infty} \hat{D}_n e^{jn\omega_0 t}$$

in which

$$|\hat{D}_n| = |D_n| \quad \text{and} \quad \angle \hat{D}_n = \angle D_n - n\omega_0 T$$

This result shows that time shifting of a periodic signal by T seconds merely changes the phase spectrum by $n\omega_0 T$. The amplitude spectrum is unchanged.

演算区

（复习）周期信号时移与时间尺度变换的傅里叶级数性质（续）

Lathi 6.3-10 | 原书第 674 页

Boulet 第4章：Properties of Continuous-Time Fourier Series，pp.139-140（Time Shifting、Time Scaling）

复习题：此前见第2批。复习第2批：重练时移与时间尺度变换对级数系数、幅相线谱及谐波频率位置的影响。已重新核对Lathi原书，完整保留左栏题干、(a)和右栏(b)及其直观解释要求；原题列出的关系式属于要求证明的结论，并非另附解答。图P6.3-9属于上一题，本题不依赖该图。

(b) Show that the exponential Fourier series for $\tilde{x}(t) = x(at)$ is given by

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn(a\omega_0)t}$$

This result shows that time compression of a periodic signal by a factor a expands its Fourier spectra along the ω axis by the same factor a. Similarly, time expansion of a periodic signal by a factor a compresses its Fourier spectra along the ω axis by the factor a. Intuitively explain this result.

演算区

（复习）周期输入经低通、高通、带通和带阻滤波器的稳态响应

Lathi 6.4-2 | 原书第 675 页

Boulet 第4章：Steady-State Response of an LTI System to a Periodic Signal，pp.155-156（含Filtering）

复习题：此前见第2批。复习第2批：把基频、傅里叶系数与各谐波经过滤波器后的稳态输出连起来。已重新核对Lathi原书，完整保留给定输入及(a)至(g)，包括最后按指定输出描述滤波器频率响应的要求。注意题目同时使用rad/s和Hz；带宽及中心频率均按原题保留。

6.4-2 A periodic signal $x(t) = 1 + 2\cos(5\pi t) + 3\sin(14\pi t)$ is applied to an LTIC system to produce output $y(t)$.

(a) Determine ω_0, the fundamental radian frequency of $x(t)$.

(b) Determine D_n, the exponential Fourier series spectrum of $x(t)$.

(c) If the system is an ideal lowpass filter with cutoff frequency $f_c = 2$ Hz, what is the output $y(t)$?

(d) If the system is an ideal highpass filter with cutoff frequency $f_c = 2$ Hz, what is the output $y(t)$?

(e) If the system is an ideal bandpass filter with a 4 Hz passband centered at 4 Hz, what is the output $y(t)$?

(f) If the system is an ideal bandstop filter with a 5 Hz stopband centered at 10 Hz, what is the output $y(t)$?

(g) Describe the frequency response of a filter that, in response to $x(t)$, would produce the output $y(t) = 4\cos(5\pi t) - 9\sin(14\pi t)$.

演算区

（复习）同一有限区间信号的不同正弦与余弦傅里叶表示

Lathi 6.5-7 | 原书第 677 页

Boulet 第4章：Determination of the Fourier Series Representation of a Continuous-Time Periodic Signal，pp.134-136；Properties of Continuous-Time Fourier Series，pp.139-140（对称性与时间反转）

复习题：此前见第2批。复习第2批：重练在指定区间上，按不同周期延拓与正弦/余弦限制建立傅里叶级数。已重新核对Lathi原书，保留(a)至(c)的基频要求，以及必要时允许直流项的原文。题目自足，无需引用相邻题的图；仅涉及三角傅里叶级数，不扩展到Legendre或Walsh等广义正交函数。

6.5-7 Represent $x(t) = t$ over the interval $(0, 1)$ by a trigonometric Fourier series that has

(a) $\omega_0 = 2\pi$ and only sine terms

(b) $\omega_0 = \pi$ and only sine terms

(c) $\omega_0 = \pi$ and only cosine terms

You may use a dc term in these series if necessary.

演算区